

KEITH TATENDA MUGWAZIWENDOTA

Telecommunications Graduate Engineer

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Portfolio: <https://keithmugwaziwendota.vercel.app>

PROFESSIONAL SUMMARY

Telecommunications engineering Graduate with 18 years of hands-on experience in network infrastructure, internet of things (IoT), artificial intelligence (AI) and embedded systems (ES). Expert in networking, active directory (AD) management, artificial intelligence (AI) driven systems and python-driven internet of things (IoT) automation. Proven track record of maintaining 98% network uptime and delivering complex technical projects ahead of schedule.

TECHNICAL SKILLS

Networking & Security: Cisco Packet Tracer, Sophos/Cisco Routers & Switches, Active Directory (AD), Windows 10 & 11, Linux, Firewall Configuration, TCP/IP, LAN/WAN, DNS.

Robotics & Embedded Systems: Arduino IDE, Raspberry Pi 4, ESP32, Proteus, Circuit Design & Simulation, PCB Design.

Programming: C++, Python, HTML, CSS, JavaScript.

Tools: Arduino IDE, Easy EDA, Visual Studio Code, Advanced Design System (ADS), Microsoft Dynamics/Business Central.

PROFESSIONAL EXPERIENCE

Hardware Specialist | AeroClock Technicals

Jan – June 2025

- ❖ **Product Reliability:** Engineered and tested PCB designs and circuit simulations, achieving 98% operational reliability for client hardware deployments.
- ❖ **Rapid Prototyping:** Streamlined the design-to-fabrication pipeline using Easy EDA and Proteus, reducing prototype iteration time by ensuring pre-fabrication accuracy.

Junior Network Infrastructure Engineer (Attachee) | National Oil Infrastructure Company of Zimbabwe (NOIC)

Aug 2023 – July 2024

- ❖ **Network Optimisation:** Managed Cisco and Sophos environments to sustain 98% uptime for critical corporate operations.
- ❖ **Infrastructure Expansion:** Successfully deployed LAN infrastructure for a new depot, expanding network capacity for 50+ new users on schedule.
- ❖ **Cybersecurity & AD:** Secured organisational data by managing user permissions in Active Directory and implementing rigorous firewall policies.
- ❖ **Technical Support:** Resolved over 2,500 technical incidents (50+ weekly) via automated ticketing, reducing average staff downtime by improving diagnostic workflows.
- ❖ **Surveillance Systems:** Monitored and maintained CCTV infrastructure, ensuring continuous security coverage and data integrity.

ACADEMIC PROJECTS

AI-Powered Wearable Assistive System (Raspberry Pi 4)

- Designed and engineered a wearable device using Raspberry Pi 4 and computer vision to assist visually impaired individuals with real-time object detection and navigation.
- Implemented Python scripts for image processing, reducing obstacle collision risks for users.

IoT Health Monitoring System (ESP32)

- Developed a remote patient monitoring system using an ESP32 microcontroller to transmit vital signs to a central server in real-time.
- Bridged the gap between hardware sensors and cloud data logging, demonstrating a practical IoT application.

Electronic Voting System (Arduino UNO)

- Programmed an automated voting unit using C++ on Arduino, featuring secure input validation and real-time tallying logic.

Rectangular Insert-Feed Rectenna (5 GHz)

- Designed and fabricated an RF energy harvesting system, optimising 5 GHz signal conversion efficiency for sustainable power solutions.

EDUCATION AND CERTIFICATIONS

Bachelor of Engineering Honours Degree in Telecommunications Engineering

Midlands State University

Class: 2.1 (Upper Second Class)

Advanced Level (ZIMSEC)

NOV 2020

Chibuwe High School

- Pure Mathematics: A
- Physics: B
- Chemistry: C

Ordinary Level (ZIMSEC)

NOV 2018

Takwirira Secondary School

- 10 Passes including Mathematics, Combined Science, and English Language.

CERTIFICATIONS

- **Cisco Networking Academy:** Networking Basics, Introduction to Cybersecurity, and Introduction to IoT.
- **Applied Machine Learning Days Africa:** Certificate of Participation (Machine Learning).
- **Disaster Ready:** Project Management Essentials and MEAL Essentials Certificate.

LANGUAGES

- English.
- Shona.

REFERENCES

Mr. G. Kanotangudza

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Mr. N Taruvinga

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